

future of seamless and visible identification for goods at least, lies in RFID and whatever technologies succeed it.

Authentication

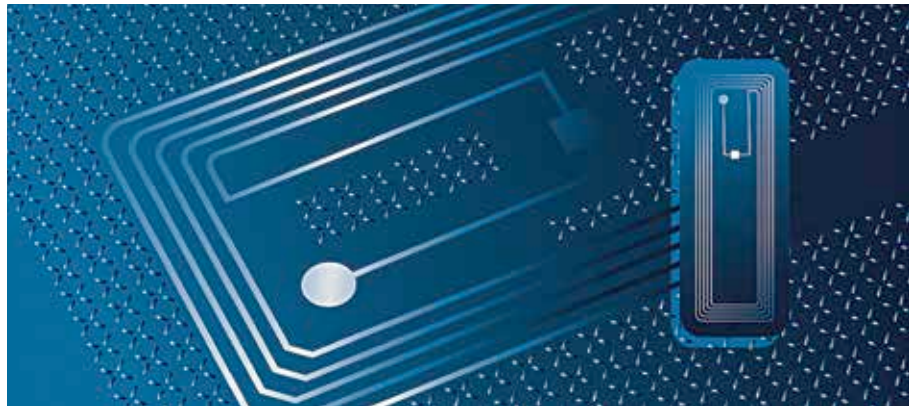
Although RFID is mainly for goods, authentication is meant to establish your identity. Currently, the most common way of doing this is via a username and password combination. Every single one of you have used authentication at some point, in fact almost all of you use it on a daily basis. Whether it's logging in to your email, your Facebook account, entering a PIN when withdrawing money at an ATM, opening your mobile bill with your unique password, swiping your finger at the biometric machine at work, unlocking your car's autocop with a button, or even opening your front door with your house key, it's all some form of authentication or the other. Again, this is not a new concept. The animal kingdom uses smells and pheromones to identify one another. This is how a mother always finds her babies, even in a herd. Cavemen probably called out "Who goes there?" in grunts, and received familiar grunts in return, and even foetuses in the womb learn the sound of their mother's voice, and after they're born are reassured when they hear it – all authentication.

The digital world deals with machines, however, and even the smartest machine ever built isn't as smart as a one hour old baby. The digital world requires a much more simplistic system of checks and balances – usernames and passwords, digital signatures, public and private key combinations, secret PINs, personalised information such as your mother's maiden name, and more.

How does authentication work?



Almost all authentication today is based on SSL certificates and the HTTPS protocol. There is no way that you can log in to gmail or icloud or any other myriad number of sites and services without it? Do you even know how it works? Head over to <http://dgigit.in/CSJuly6> to find out more about this and RFID tagging and other related information.



A miniature version of this might be all that you'll ever need for authentication

It's easier to have a hit and miss, yes or no, black or white system, but things get complicated in the technology world. For example, only you should have full access to your PC to install and uninstall software, but you still want others to be able to use it to surf and check email. A database system might need a group of people to be able to read data, another to also be able to write data, and a third to be able to read, write and delete data. And these are still the most simplistic scenarios.

The problem with digital authentication thus far has been us. PEBKAC – Problem Exists Between Keyboard And Chair – is our favourite way of highlighting that the human element is often the weakest link in most technology systems. Sharing your password with people, using weak passwords, or worse using even weaker password reset questions all lead to a very unreliable digital authentication experience. As dumb as humans are, we're also the designers of systems, and this means that we're also the first to outsmart ourselves. For every lazily set weak password, there's a hacker or his bots probing for access. Even seemingly fool-proof technologies such as Captcha (images used to verify that you're human and not a bot) are easily fallible. In fact, the username-password form of digital authentication is perhaps the worst and weakest of all, and yet is the most commonly used.

Thankfully, biometrics has brought about a more reliable method of proving you are indeed you. Credit cards with photographs try and add an additional layer of authentication – signatures are no good, and we confirmed this in an article we did a few years ago when we signed for purchases as Batman or Mickey Mouse, or

something. Fingerprint scanners can't be used en-masse due to the lack of a central database of them, and the privacy concerns that people have. So what are we left with? Are we doomed to forever have to invent weirder and stronger passwords?

Perhaps not. There's already talk of using unique digital tattoos to safeguard your identity, and bordering on sci-fi is the research that's being done into making unique identifiers that are built into pills – each of these pills contains a unique RFID identifier, and is powered by the digestive enzymes in your stomach. In essence, such a technology would make your entire body an RFID transmitter. However, what if you forgot to take your pill one day, or what if you ran out of pills on Saturday night? Would you spend Sunday doing nothing because no system you wanted to use would allow you to, and of course the pill delivery guy decided to spend his Sunday with his family? There are still chinks to be worked out, but it's a step in the right direction at least.

Automation

If there's one thing technology is great at, it's automation. The drudgery of repeating a task hundreds of times can bore humans to sleep, but that's exactly where technology excels. From simple calculators that replaced log tables and complicated calculations, usually for the purposes of accounting, to repetitive tasks done in manufacturing which are now being done by robots, technology is removing the drudgery from jobs. Of course, what this usually means is that people lose boring jobs, but it also creates more exciting jobs – designing such automated systems, for example. Another thing technology has going for it is speed, and though in some